

RIPHAH INTERNATIONAL UNIVERSITY
Islamabad

Faculty of Computing
BS Cyber Security
Course: Cy4-1 Operating System

USER INTERFACE THEME
Project Report

Submitted By
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Submitted To
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Declaration

It is hereby declared that the project titled

"User Interface Theme" submitted in partial fulfillment of the requirements for the course **Operating System (Cy4-1)** is an original work carried out under the supervision of **Mr. Yawar Abbas**. This work has not been submitted previously for any degree, diploma, or any other course in any institution.

Raja Ali Shahid

SAP ID: 63158

Certificate

This is to certify that the project report entitled **"User Interface Theme"**, submitted by **Raja Ali Shahid** (SAP ID: **63158**), in partial fulfillment of the requirements for the course **Cy4-1 Operating System**, Riphah International University, is a record of bona fide work carried out under my supervision and guidance.

Mr. Yawar Abbas

Project Guide

Faculty of Computing

Riphah International University, Islamabad

Acknowledgments

I am thankful for the opportunity to work on the project "User Interface Theme". I sincerely appreciate the guidance and support provided by my project supervisor

Mr. Yawar Abbas during the development of this project.

This project has been developed using Python and implemented in VS Code.

I am also grateful to my teachers, friends, and family for their continuous support and encouragement.

Raja Ali Shahid

1. Executive Summary

This report provides a detailed analysis of the

User Interface Theme project, a Python-based application developed to enhance and customize the visual appearance of user interfaces. The project focuses on improving design elements such as layout, colors, and overall user experience to make applications more attractive and user-friendly.

This project has been developed individually under the academic requirements of the course Operating System (Cy4-1) at Riphah International University. It also includes additional features such as camera access, multi-factor authentication, and a blur camera effect to enhance security and user interaction.

This document is intended for academic evaluation and covers the project's objectives, methodology, design approach, and implemented features.

1.2 Overview

My project is titled "**User Interface Theme**". It is a Python-based application developed in **VS Code**. The main purpose of this project is to improve the visual design and user experience of software applications.

In this project, I focused on customizing the user interface by changing elements like colors, layout, and overall theme design so that the application looks more attractive and user-friendly.

Additionally, I have included features like **camera access, multi-factor authentication, and blur effect on camera** to improve both usability and security.

Project Objectives

The **User Interface Theme** project aims to:

- Improve the visual appearance of applications by customizing colors, layouts, and overall design.
- Enhance user experience by creating a more interactive and user-friendly interface.
- Provide theme customization options to make the interface more flexible and attractive.
- Implement additional features such as camera access, multi-factor authentication, and blur camera effect to improve security and usability.
- Ensure smooth performance and responsiveness of the user interface.
- Follow good design principles to make the application easy to use and understand.

1.3 Literature Review

Research shows that an effective User Interface (UI) plays a crucial role in the success of any software application. Studies highlight that good design, appropriate color selection, and well-structured layouts significantly improve user experience.

In GUI (Graphical User Interface) development, modern tools and libraries are appreciated for creating intuitive and user-friendly interfaces, especially Python-based UI frameworks.

Additionally, recent studies emphasize the importance of user privacy and data security in application design.

This project applies these principles by developing a User Interface Theme that focuses on improved design, usability, and user interaction. It also incorporates features such as multi-factor authentication and camera-based functionality to enhance both security and overall user experience.

5. System Architecture

The architecture of the User Interface Theme project includes the following modules:

Module	Description
□ GUI Layer	A Python-based user interface that manages user interaction and displays different themes, layouts, and design elements.
□ Theme Management Module	Handles customization of colors, styles, fonts, and overall layout of the interface.
□ User Interaction Module	Manages user inputs and ensures smooth navigation and responsiveness within the application.
□ Authentication Module	Implements multi-factor authentication to enhance security.
□ Camera Module	Provides camera access with blur effect for improved privacy and user experience.
□ Development Environment	The project is developed using Python in VS Code.

8. Implementation Details

The User Interface Theme project was implemented using **Python in VS Code**, focusing on creating a visually appealing and user-friendly interface.

The application was developed using a GUI framework where different UI components such as buttons, labels, input fields, and layouts were designed and customized. Various themes were applied by modifying colors, fonts, and styles to improve the overall look and feel of the application.

A **theme management system** was implemented to allow dynamic switching between different UI styles. This enhances flexibility and provides a better user experience.

For security enhancement, a **multi-factor authentication system** was integrated. This ensures that only authorized users can access the application.

Additionally, a **camera module** was implemented that allows camera access and applies a **blur effect** for privacy and better visual interaction.

0. Demonstration Analysis

Login Screen

This figure shows the login screen of the User Interface Theme application. It serves as the entry point where the user enters credentials to access the system.

Multi-Factor Authentication Login Screen

This figure shows the secure login interface of the User Interface Theme application. It uses a QR code and OTP-based verification to enhance user authentication and security.

User Dashboard with Camera Access and Blur Feature

This figure shows the main dashboard of the User Interface Theme application after successful login. It includes camera access functionality with a blur effect to enhance privacy and user security.

Secure Login System - Face Recognition Interface

This interface shows an admin dashboard of a secure login system with live camera access for face detection. It allows starting/stopping the camera and enabling or disabling face blur for privacy control.

Dark Mode

The application implements a sleek Dark Mode interface designed to reduce eye strain and improve visual clarity, especially during extended usage in low-light environments. By utilizing a high-contrast dark color scheme, the UI enhances readability and provides a modern, professional aesthetic while optimizing power efficiency for the user.

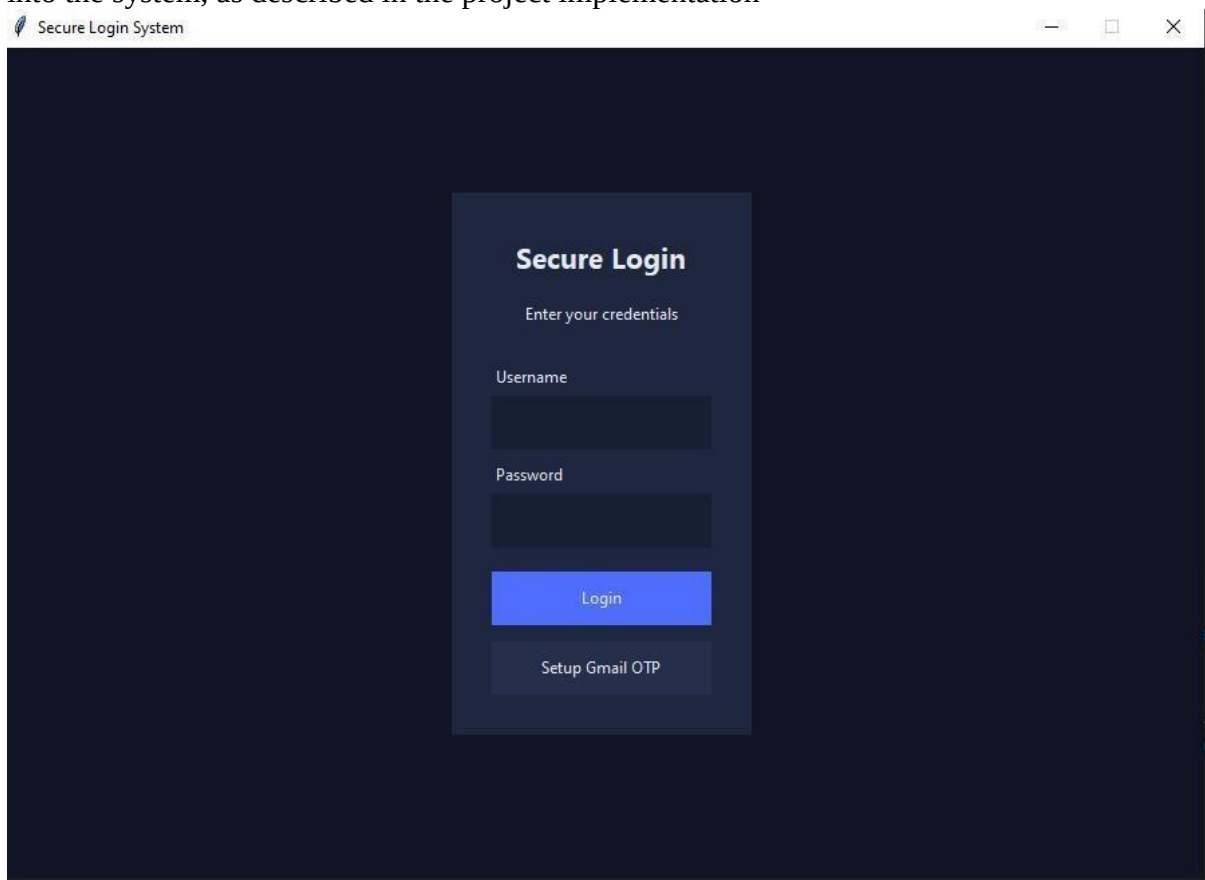
Light Mode

The Light Mode interface provides a clean and classic aesthetic, utilizing a bright color palette to ensure maximum legibility and professional appeal in well-lit environments. This theme focuses on high-contrast text against a neutral background, offering a familiar user experience that prioritizes clarity and a minimal design approach.

1. Login Interface

The login interface represents the entry point of the Secure Login System where users enter their credentials to access the application. It includes input fields for username and password along with a login button.

The design follows a clean and minimal layout using a dark theme to improve readability and user focus. An additional option for setting up Gmail OTP is also provided, enabling multi factor authentication for improved security. This screen ensures that only authorized users can proceed into the system, as described in the project implementation

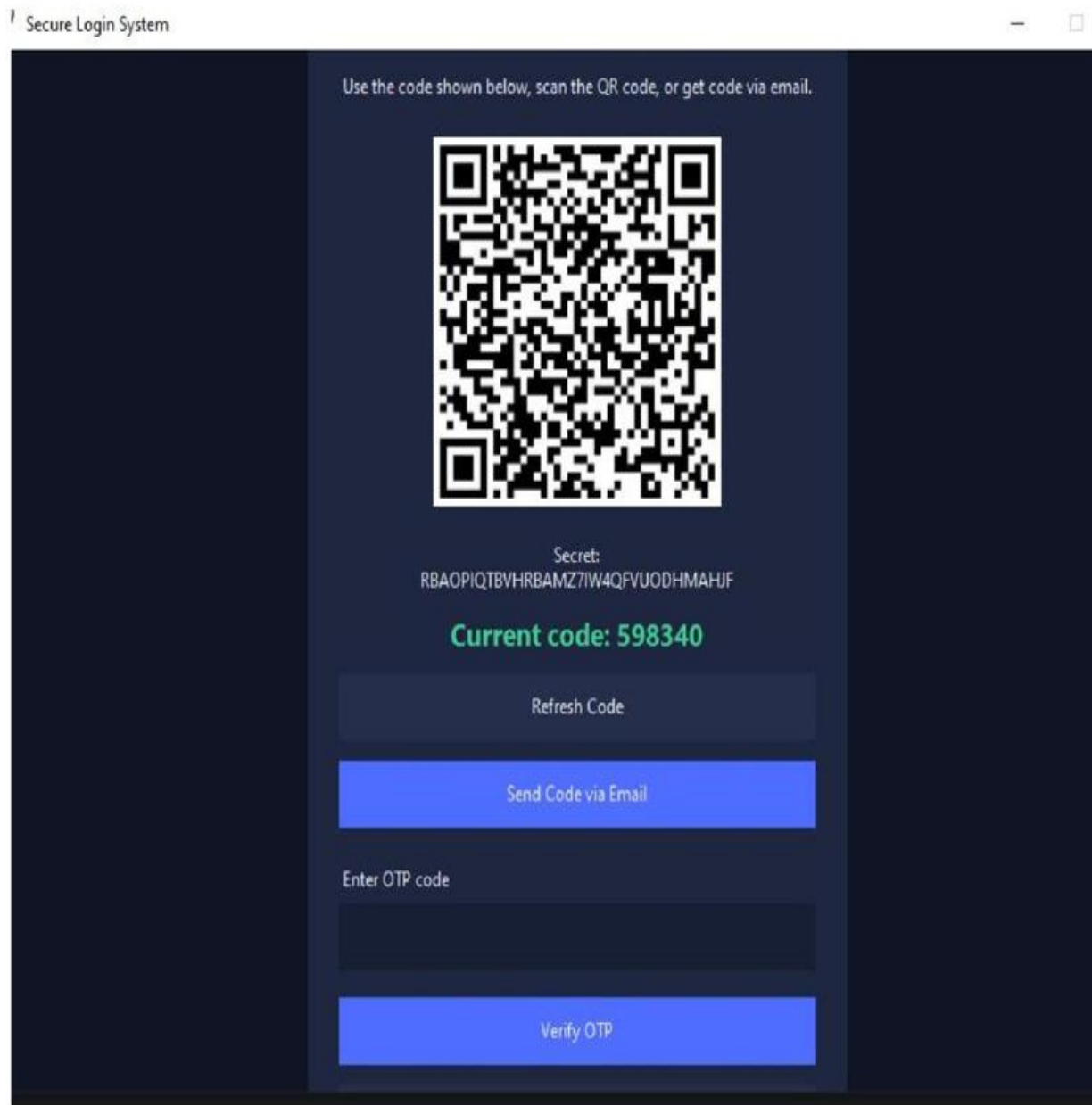


2. Multi Factor Authentication (OTP and QR Code)

This interface demonstrates the multi factor authentication process used to enhance system security. It provides a QR code and a time based OTP for user verification.

Users can scan the QR code or receive the OTP via email to complete the authentication process. This additional verification step protects the system from unauthorized access.

According to the report, this authentication module strengthens security by integrating OTP based verification mechanisms



User Dashboard with Camera Access and Blur Feature

3. User Dashboard with Camera Access

The dashboard is displayed after successful login and provides access to the main functionalities of the system. It includes controls for managing the camera and interacting with system features.

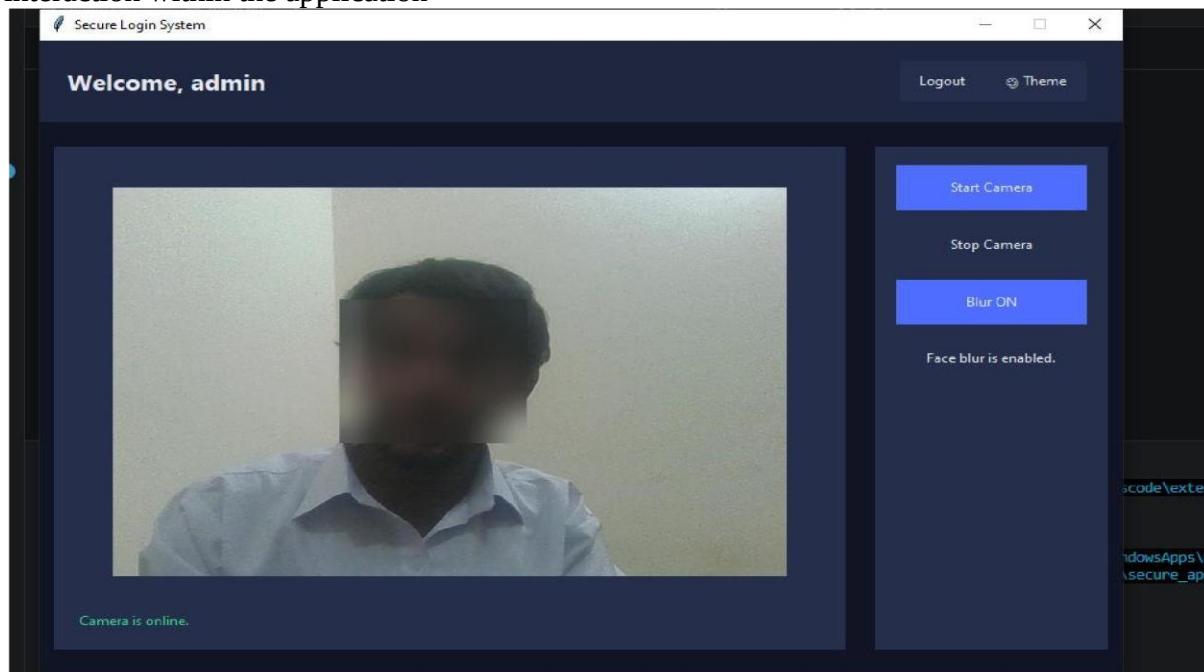
Users can start and stop the camera, enabling real time monitoring through the interface. The layout is structured to ensure smooth navigation and ease of use.

4. This component is part of the GUI layer responsible for handling user interaction and system operations Face Blur Feature (Privacy Control)

This section highlights the face blur functionality integrated within the camera module. It allows users to enable or disable face blurring for privacy protection.

When enabled, the system automatically obscures the user's face in the camera feed. When disabled, the original video feed is shown without modification.

The report confirms that this feature is implemented to enhance user privacy and improve interaction within the application



Secure Login System – Face Recognition Interface

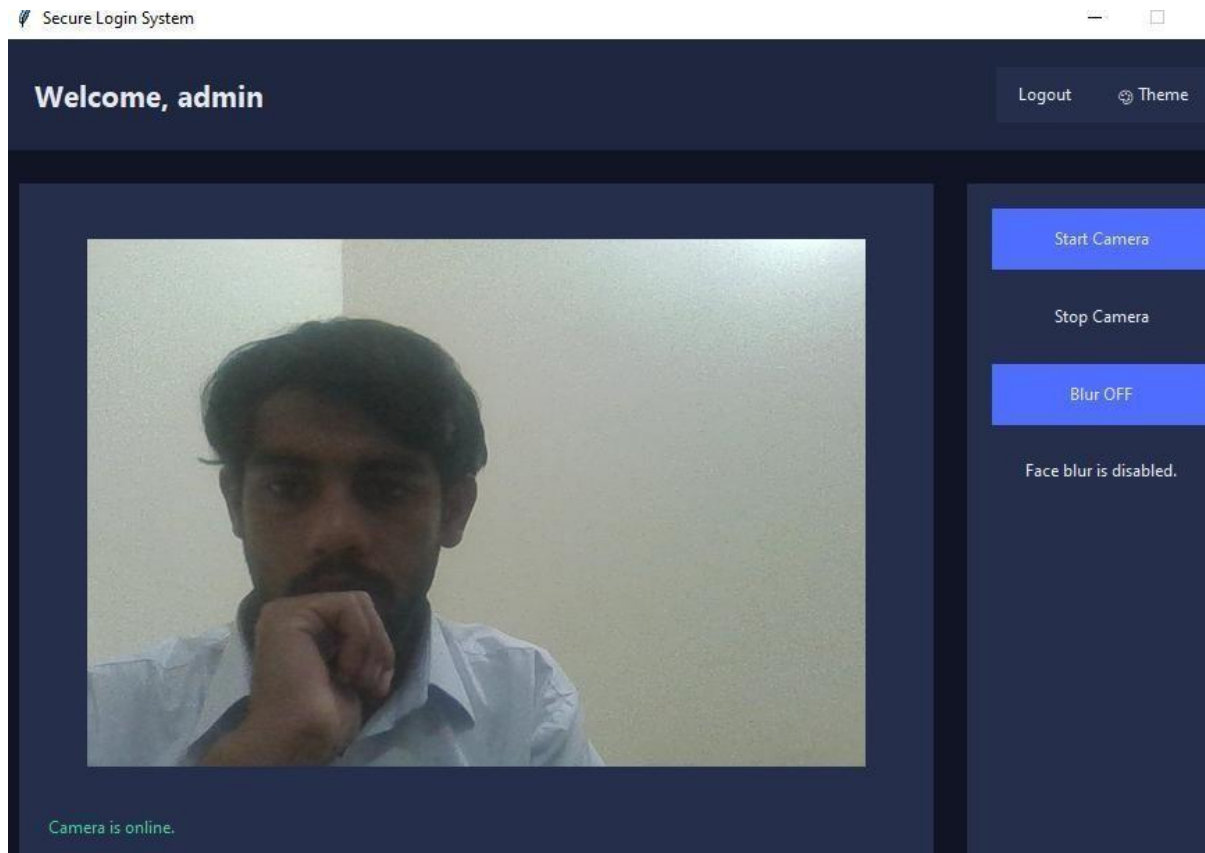
This interface shows an admin dashboard of a secure login system with live camera access for face detection. It allows starting/stopping the camera and enabling or disabling face blur for privacy control.

5. Dark Mode Interface

The dark mode interface is designed to reduce eye strain and improve visibility, especially in low light environments. It uses a high contrast color scheme to enhance readability.

This mode provides a modern and professional appearance while maintaining clarity in the user interface. It is particularly useful for extended usage.

As mentioned in the report, dark mode improves both visual comfort and overall user experience



Dark mode:

6. Light Mode and Theme Customization

The light mode interface offers a bright and clean design suitable for well lit environments. It focuses on clarity and simplicity, making the interface easy to read and use.

In addition to light mode, the system provides theme customization options. Users can modify colors such as background, buttons, and text according to their preferences.

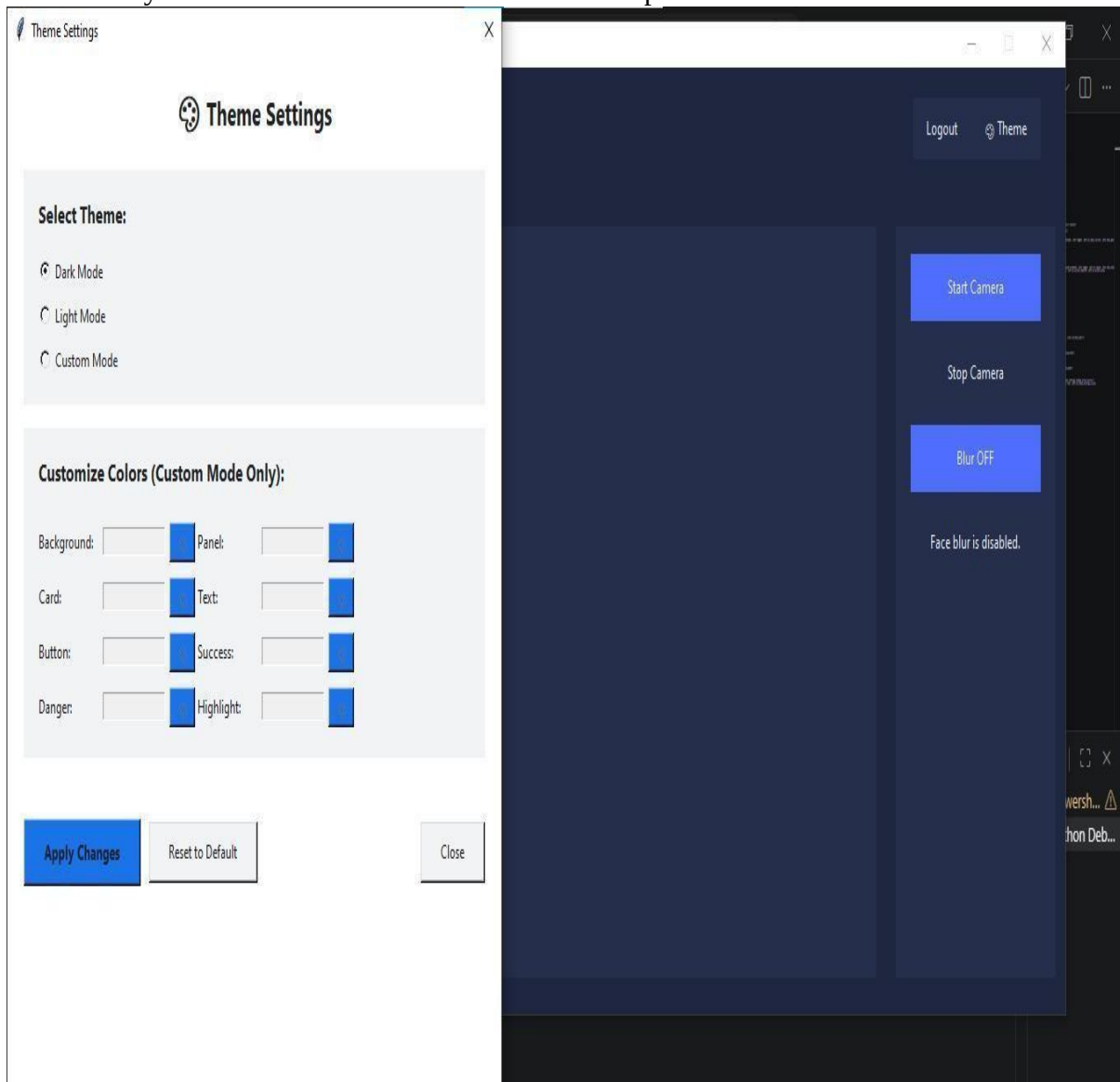
The theme management system allows dynamic switching between different styles, increasing flexibility and usability .

7. Overall System Interface

The overall system integrates authentication, dashboard controls, camera functionality, and theme customization into a unified application.

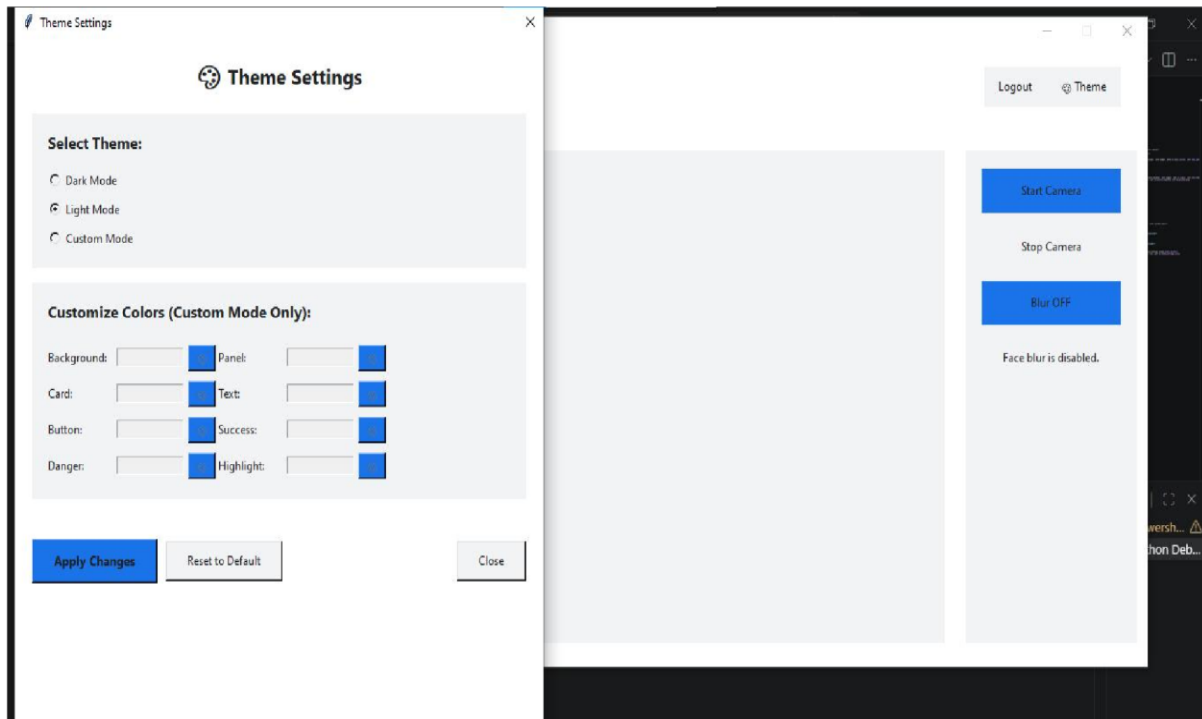
The design follows a modular approach, where each component such as authentication, user interface, and camera module works independently yet collaboratively.

According to the project conclusion, the system successfully combines usability and functionality to deliver a smooth and efficient user experience



The **Light Mode interface** provides a clean and classic aesthetic, utilizing a bright color palette to ensure maximum legibility and professional appeal in well-lit environments."

"This theme focuses on high-contrast text against a neutral background, offering a familiar user experience that prioritizes clarity and a minimal design approach."



Conclusion

The project's user interface is based on a simple, clean, and user-friendly theme.

Its design is modular, making it easy to understand and maintain.

The interface provides users with a smooth and efficient experience.

Overall, the project combines usability and functionality to present an effective solution.

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